👩‍🏫 Staff Guide · Biology A · Unit 3 Biofuel & Energy (Level 1)

Level 1 = about 2nd-grade reading, with emoji and pictures. SLE students stay in Ms. Dunham's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Unit 2 Algae Biofuel: A3 Matter & Energy (Characteristics of Matter and Energy, Energy Transfer & Transformation, Crash Course Chemistry, Bill Nye Energy), A5 Making Lipids, A6 Making Biofuel + Combustion Lab, A7 Energy Reactions, A8 Carbon Cycle, A10 Global Connections, A2.10 Algae Biofuel Review, Algae Biofuel Final Exam")	**Level 1 piece**
A3 Matter & Energy worksheets + Crash Course Chemistry + Bill Nye Energy	Notes part A + slides 1-6 (her two A3 worksheets grouped as one lesson)
A5 Making Lipids (Lego model)	Notes part B + lipid slide
A6 Making Biofuel + Combustion Lab (10 pts)	Biodiesel observer packet (🔬 Labs & Projects)
A7 Energy Reactions	Energy-changes packet + notes part D
A8 Carbon Cycle (+ webquest)	Notes part E + carbon-cycle order game
A10 Global Connections	🚧 Under construction (her topics are only pictures)
Algae Test review + Algae Biofuel Final Exam / paper Algae Test	Modified unit test + key
Class slides	Deck: 20 slides, Words to Know first, "point to the picture" checks, recap
Her worksheets (grouped into lessons)	Guided notes: 15 items + key
Class quiz / notes check	Quiz: 8 questions + key
Vocabulary	13 picture cards (cut apart)
—	2 games on the class site: matching + step-by-step order

2. Supports built in

- **Read aloud** every line; student may point, say, or circle (point = full answer)
- One idea per slide; big pictures; 2–3 picture choices
- Picture anchors: ⚡ energy = makes things move, 🪨 stored vs 🏃 moving, 🧈 lipid = oil
- Level 1 uses no chemical formulas; Level 2 uses word equations only

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part
- Cover one choice to make it 2

4. What to grade

- Unit test: same weight as the class Algae test, scaled
- **Notes**: same points as her worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver, Room 113 (SLE): changes and IEP questions.
- Facts we could not confirm in her files say "check with Ms. Dunham".

🚧 Under construction (check with Ms. Dunham)

- **A10 Global Connections**: Her 24 connection topics are only in a picture PDF (no text). We need the topic list to build a picture reading.
- **Carbon cycle webquest**: Title only in Canvas; no file or link.
- **Biofuel sorter**: Title only in Canvas; no file.
- **Paper Algae Test (32 pts)**: Not in the export. Our test is built from her Final Exam quiz and review.

6. Photo credits (open licenses)

Chlorella vulgaris NIES2170.jpg — ja:User:NEON / User:NEON_ja, CC BY-SA 3.0, Wikimedia Commons · CSIRO ScienceImage 3977 Canola Flowers.jpg — Carl Davies, CSIRO, CC BY 3.0, Wikimedia Commons · Biodiesel.JPG — Shizhao拍摄, CC BY-SA 3.0, Wikimedia Commons · Biofuel pumps DCA 07 2010 9834.JPG — Mariordo Mario Roberto Duran Ortiz, CC BY-SA 3.0, Wikimedia Commons

👩‍🏫 Staff Guide · Making & Burning Biodiesel (Observer Jobs) (Level 1)

Biology A · Unit 3 · goes with the class “A6 Making Biofuel, Day 2 Burning Biofuel, Combustion Lab report” (5 + 5 pts). Level 1 = about 2nd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Observer card**: A6 Making Biofuel + Day 2 burning + Combustion Lab: our students observe and record; staff or the teacher handle methanol, the catalyst and the flame. 6 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “What color is the oil?”
- “Is the water hotter or colder now?”
- “Point to the top layer.”

3. Modifications

- Observer job only (no chemicals, no flame)
- Circle-the-picture data instead of numbers
- Point/say answers count fully

4. What to grade (5 + 5 pts)

- Same 5 + 5 points: did my job, data circles done, goggles on
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

🚧 Under construction (check with Ms. Dunham)

- **This packet**: Ms. Dunham said in Canvas the teacher adds the catalyst. Please confirm our students' observer roles and where the safety line goes.

6. Photo credits (open licenses)

Biodiesel.JPG — Shizhao拍摄, CC BY-SA 3.0, Wikimedia Commons

👩🏾‍🏫 Staff Guide · Energy Changes: Hot or Cold? (Level 1)

Biology A · Unit 3 · goes with the class “A7 Energy Reactions” (5 pts). Level 1 = about 2nd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Lab card**: A7 Energy Reactions: Investigation 1 (baking soda + vinegar) and Investigation 2 (baking soda solution + calcium chloride). 6 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Is the number going up or down?”
- “Hot or cold?”
- “Which one is like a hand warmer?”

3. Modifications

- Hot/cold picture choice instead of numbers
- Staff reads the thermometer with the student
- No energy-arrow drawing

4. What to grade (5 pts)

- Same 5 points: both cups circled, job done
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

⛽ Biofuel & Energy Notes

Biology A · Unit 3 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.

⚡ energy _____

🧈 oil _____

🔥 heat _____

🌿 algae _____

🖼️ B. Fill in the word. Use the pictures.

5. 🏃 Energy makes things _____.

6. 🧈 Algae make _____.

7. 🔥 Burning fuel makes _____.

8. ♻️ Algae fuel can grow _____.

Word bank: ♻️ **back** | 🔥 **heat** | 🏃 **move** | 🧈 **oil**

⭕ C. Circle the right picture.

9. Stored energy:


Kid running


Fuel in a tank

10. This gets hot:


Cold pack


Hand warmer

11. This fuel grows back:


Algae


Coal

🔗 D. Draw a line from the picture to the word.









heat

food energy

biodiesel

light

📖 Biofuel & Energy: Vocabulary Picture Cards (Level 1)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Energy

makes things move or change



Matter

stuff you can touch



Potential energy

stored energy, waiting



Kinetic energy

moving energy

Chemical energy

energy stored in food and fuel

Thermal energy

heat



Lipid

fat or oil



Biodiesel

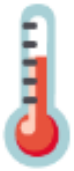
fuel made from plant oil

Renewable

can grow back fast

Fossil fuel

old fuel from deep in the ground



Exothermic

gets hot



Endothermic

gets cold



Carbon cycle

carbon goes around and around

🛢️ Making & Burning Biodiesel (Observer Jobs)

Biology A · Unit 3 · Level 1



Name: _____

A6 Making Biofuel + Day 2 burning + Combustion Lab: our students observe and record; staff or the teacher handle methanol, the catalyst and the flame.

🧺 What I need (observer): get it. Check ☐.

- ☐ 🥽 **1** pair of goggles (on the whole time)
- ☐ 📋 **1** observer card + pencil
- ☐ ⏱️ **1** timer or clock
- ☐ 📷 **1** tablet for photos (if allowed)

🌱 Do one box. Check ☐ it.

1. 🌻 Look at the **oil**. Say the color. ☐
2. ⏱️ Teacher shakes it. I time **10 minutes**. ☐
3. 🥞 Next day: see **2 layers**. ☐
4. 🔥 Teacher burns the fuel. I **watch** from my spot. ☐
5. 🌡️ The water gets **hot**. I say the number. ☐
6. 📷 I take a photo or check ☐. ☐

📋 My data. Circle or write.

Oil color: 🟡 yellow 🟠 orange ⚪ clear

2 layers? ✅ yes ❌ no

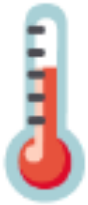
Water got: 🔥 hotter 🧊 colder

💡 *This lab is for watching and recording. Our students never handle methanol, potassium hydroxide, or the flame.*

🚧 **Under construction:** Ms. Dunham said in Canvas the teacher adds the catalyst. Please confirm our students' observer roles and where the safety line goes. *Check with Ms. Dunham.*

🌡️ Energy Changes: Hot or Cold?

Biology A · Unit 3 · Level 1



Name: _____

A7 Energy Reactions: Investigation 1 (baking soda + vinegar) and Investigation 2 (baking soda solution + calcium chloride).

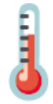
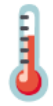
🧺 Investigation 1: get it. Check ☐.

- ☐  **1 cup** vinegar
- ☐  **1 cup** baking soda
- ☐  **1** thermometer

🧺 Investigation 2: get it. Check ☐.

- ☐  **1 cup** baking soda solution
- ☐  **1 cup** calcium chloride (staff pours)
- ☐  **1** thermometer

🌱 Do one box. Check ☐ it.

1		Put the **thermometer** in the cup. Say the number. ☐
2		Add the **baking soda**. Fizz! ☐
3		**Watch** the number. ☐
4		It went **down**: cold! ☐
5		Cup 2: staff adds the white powder. ☐
6		It went **up**: hot! ☐

🌡️ Circle what happened.

Cup 1 (vinegar): 🔥 hot 🧊 cold

Cup 2 (white powder): 🔥 hot 🧊 cold

🛢️ Making & Burning Biodiesel (Observer Jobs) · Get Ready

Biology A · Unit 3 · Level 1



Name: _____

👥 My job. Circle it.

⏱️ **Timer** I watch the clock.	📝 **Recorder** I write it down.	📷 **Photo taker** I take pictures.	👀 **Color checker** I say the color.

🧰 Get the tools. Check ☐.

- ☐ 🥽 Goggles
- ☐ ⏱️ Timer
- ☐ 🌡️ Thermometer
- ☐ 📋 Data table

⚠️ Be safe!

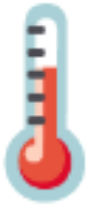
- 🥽 Goggles on!
- 🧪 Do not touch the bottles.
- 🔥 Stay back from fire.
- 🧼 Wash hands.

✅ Before we start

☐ Goggles on ☐ I know my job ☐ Safety line on the floor ☐ Hands free (no food)

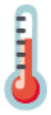
🌡️ Energy Changes: Hot or Cold? · Get Ready

Biology A · Unit 3 · Level 1



Name: _____

👥 My job. Circle it.

 **Reader** I read the number.	 **Adder** I pour it in.	 **Recorder** I write it.	 **Cleaner** I clean.

🧰 Get the tools. Check ☐.

- ☐  Thermometer
- ☐  Goggles
- ☐  Gloves
- ☐  Paper towels

⚠️ Be safe!

- 🥽 Goggles on!
- 🧤 Do not touch the white stuff.
- 🚫 No tasting.
- 🧼 Wash hands.

✅ Before we start

☐ Goggles on ☐ Thermometer in the cup ☐ Start number written

🛢️ Making & Burning Biodiesel (Observer Jobs) · My Check

Biology A · Unit 3 · Level 1



Name: _____

📸 My photos. Check ☐.

- ☐ 🌻 The oil
- ☐ 🥞 2 layers
- ☐ 🔥 The flame (safe spot)

✅ I did it! Check ☐.

- ☐ 👀 I watched safely
- ☐ 📝 I wrote or said what I saw
- ☐ 🤝 I did my job
- ☐ 🥽 I wore goggles

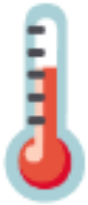
⭕ Circle one face.

🥽 **Did I stay safe?**	😀	🙂	😕
🤝 **Did I do my job?**	😀	🙂	😕
🌡️ **Did I see the water get hot?**	😀	🙂	😕

⭐ I learned ____. Draw or write.

🌡️ Energy Changes: Hot or Cold? · My Check

Biology A · Unit 3 · Level 1



Name: _____

📸 My photos. Check ☐.

- ☐  Cup 1 fizzing
- ☐  My thermometer

✅ I did it! Check ☐.

- ☐  I read the thermometer
- ☐  I said hot or cold
- ☐  I did my job
- ☐  I was safe

⭕ Circle one face.

 **Did I read the number?**	😀	🙂	😕
 **Did I do my job?**	😀	🙂	😕
 **Was I safe?**	😀	🙂	😕

⭐ I learned ____. Draw or write.

🚧 Biofuel Sorter

Biology A · Unit 3 · Level 1 · Under construction

Name: _____

🚧 **Under construction:** Canvas has only the title. We do not know what students sort. *Check with Ms. Dunham.*

What we have so far

The matching game sorts renewable and non-renewable fuels for now.

Staff: until this is ready

Have the student do the matching Level 1 unit game or vocabulary cards for this lesson, and give the same points for effort. Mr. Driver, Room 113.

✅ Biofuel & Energy Quiz

Biology A · Unit 3 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. 🍎 Living things use energy from…

-  Rocks
-  Food

2. 🧱 Matter is stuff you can…

-  Touch
-  Only think

3. 🔥 Burning fuel makes…

-  Heat
-  Ice

4. 🧈 Algae make…

-  Milk
-  Oil

5. ♻️ Which fuel grows back?

-  Coal
-  Corn fuel

6. 🌡️ **Hot reaction: the number goes…**

- ⬆️ Up
- ⬇️ Down

7. 🌿 **Who takes carbon OUT of the air?**

- 🔥 Fire
- 🌿 Plants and algae

8. 🐣 **Good fuel algae make… babies.**

- ➕ Lots of
- ➖ Few

🚧 Carbon Cycle Webquest

Biology A · Unit 3 · Level 1 · Under construction

Name: _____

🚧 **Under construction:** Canvas has only the title. There is no file or website link. *Check with Ms. Dunham.*

What we have so far

Use the carbon-cycle order game and notes part E for now.

Staff: until this is ready

Have the student do the matching Level 1 unit game or vocabulary cards for this lesson, and give the same points for effort. Mr. Driver, Room 113.

🚧 A10 Global Connections

Biology A · Unit 3 · Level 1 · Under construction

Name: _____

🚧 **Under construction:** Her 24 Global Connections topics are only pictures in the export (no words). We need the list of topics to make the picture reading. *Check with Ms. Dunham.*

What we have so far

The worksheet frame: for each connection write the title, what it is, and how I can help.

Staff: until this is ready

Have the student do the matching Level 1 unit game or vocabulary cards for this lesson, and give the same points for effort. Mr. Driver, Room 113.

⛽ Biofuel & Energy Notes · KEY

Biology A · Unit 3 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.

⚡ energy _____

🧈 oil _____

🔥 heat _____

🌿 algae _____

🖼️ B. Fill in the word. Use the pictures.

5. 🏃 Energy makes things **move**.

6. 🧈 Algae make **oil**.

7. 🔥 Burning fuel makes **heat**.

8. ♻️ Algae fuel can grow **back**.

Word bank: ♻️ **back** | 🔥 **heat** | 🏃 **move** | 🧈 **oil**

⭕ C. Circle the right picture.

9. Stored energy:

- Kid running
-  Fuel in a tank

10. This gets hot:

-  Cold pack
-  Hand warmer

11. This fuel grows back:

-  Algae
-  Coal

🔗 D. Draw a line from the picture to the word.

Key: 💡 → light · 🔥 → heat · 🍎 → food energy · 🛢️ → biodiesel

	heat
	food energy
	biodiesel
	light

🔑 *Key written by our team from Ms. Dunham's materials, please check.*

✅ Biofuel & Energy Quiz · KEY

Biology A · Unit 3 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. 🍎 Living things use energy from… Gen ed #16

-  Rocks
-  Food

2. 🧱 Matter is stuff you can… Gen ed #10

-  Touch
-  Only think

3. 🔥 Burning fuel makes… Gen ed #22

-  Heat
-  Ice

4. 🧈 Algae make… From review

-  Milk
-  Oil

5. ♻️ Which fuel grows back? Gen ed #26

-  Coal
-  Corn fuel

6. 🌡️ Hot reaction: the number goes… Gen ed #24

- ✅ ⬆️ Up
- ⬇️ Down

7. 🌿 Who takes carbon OUT of the air? Gen ed #27

- 🔥 Fire
- ✅ 🌿 Plants and algae

8. 🐣 Good fuel algae make… babies. Gen ed #1

- ✅ ➕ Lots of
- ➖ Few

Answers

1-B (Food) · 2-A (Touch) · 3-A (Heat) · 4-B (Oil) · 5-B (Corn fuel) · 6-A (Up) · 7-B (Plants and algae) · 8-A (Lots of)

Level 1 version of the gen ed "Algae Biofuel Final Exam (Canvas quiz) + A2.10 Algae Biofuel Review". Read every question aloud. Score: __ / 8.

🔑 *Questions tagged "Gen ed #" use the answers stored in her Canvas Algae Biofuel Final Exam. "From review" = written by our team from her A2.10 review, please check.*